🧪 Science Starts: Safety & Lab Skills: Reading · ANSWER KEY

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

How Scientists Work Safely

Every science lab starts with safety. At Ferndale, students sign the science safety contract. Goggles go on any time there are chemicals, heat, or glass. There is no food, drink, or gum in the lab. Long hair is tied back, shoes cover the whole foot, and loose sleeves and jewelry are secured.

Accidents must be reported to the teacher right away, even small ones. If someone is hurt, a student yells “Code one!” so the teacher comes fast. Following every direction, and asking when you are not sure, keeps everyone safe.

Scientists start with observations. An observation uses the five senses to describe what is really there: “The lollipop is red and round.” An inference is what you think it means: “It will taste like cherry.” Good science keeps the two apart.

Next, a scientist writes a hypothesis, a testable prediction. A strong hypothesis uses If… then… because…: “If I suck on the lollipop without biting, then it will dissolve in 8 minutes, because it is small.”

In an experiment, the independent variable is what the scientist changes. The dependent variable is what the scientist measures. Everything else stays the same. After collecting data, the scientist writes a conclusion: was the hypothesis supported or not supported, and what data shows it?

Check your reading

1. Name two things you must do to dress safely for lab.
Any two: tie back long hair, wear closed-toe shoes, secure loose sleeves, take off dangling jewelry.

2. What is the difference between an observation and an inference?
An observation uses the five senses to describe what is there; an inference is what you think it means.

3. In the lollipop lab, what is the dependent variable?
The time it takes the lollipop to dissolve (what is measured).